CAPSTONE PROJECT SCOPE DOCUMENT

Project: KijaniKiosk End-to-End Delivery
Track Selected: Track A – Infrastructure-First

Problem Statement:
KijaniKiosk currently operates with basic infrastructure and limited production-readiness. The goal is to extend the system into a production-approaching, multi-environment, monitored deployment with automated CI/CD and environment isolation.

In-Scope Components:
- Provisioning a dedicated 'kijani-staging' namespace using Terraform.
- Configuring environment separation via Ansible.
- Deploying kk-payments to both staging and production using the same Deployment manifest.
- Creating environment-specific ConfigMaps (DB_HOST differences).
- Implementing a Jenkins pipeline with automated staging deploy, smoke tests, and a manual approval gate before production deployment.
- Integrating the Week 10 serverless receipt chain with staging.
- Implementing Prometheus alerting (restart count, error rate, or latency).

Out-of-Scope:
- Refactoring the application source code.
- Implementing new microservices beyond kk-payments.
- Full production-grade autoscaling or service mesh.
- Advanced monitoring dashboards (Grafana not required).

Success Criteria:
- A new engineer can clone the repo and deploy the system within one day.
- Terraform successfully provisions dedicated staging infrastructure.
- Jenkins pipeline runs end-to-end with approval gate clearly visible.
- kk-payments deploys successfully to both environments using the same manifests.
- Prometheus alert rule fires correctly under simulated failure.
- Receipt chain triggers end-to-end from staging kk-payments.

Architecture Diagram:
(Delivered as a separate PNG file.)